

do all of these questions by hand, i.e. without a computer. of course, check your answers with Julia.

orthonormal bases

1. find two orthogonal vectors in the plane $x + y + 2z = 0$. make them orthonormal.
2. if the matrix Q has orthonormal columns, what is the least squares solution $\hat{\mathbf{x}}$ to $Q\mathbf{x} = \mathbf{b}$?
3. (a) find orthonormal vectors $\mathbf{q}_1, \mathbf{q}_2$ that span the column space of:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix} \quad (1)$$

- (b) write the $Q = [\mathbf{q}_1, \mathbf{q}_2]$ and R matrices for the factorization $A = QR$. (check that it works in Julia by $Q * R$. note $Q, R = \text{qr}(A)$ should give the same directions but maybe different signs.)

- (c) faced with the overdetermined system $A\mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 7 \end{bmatrix}$, use your $A = QR$ factorization to find the least squares solution $\hat{\mathbf{x}}$. you should arrive at a 2 by 2 triangular system to solve for $\hat{\mathbf{x}}$. (check your answer with $A \setminus \mathbf{b}$ in Julia.)

- (d) finally, find a third vector $\mathbf{q}_3$ that makes $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ an orthonormal basis for $\mathbb{R}^3$. which of the four fundamental subspaces of A contains $\mathbf{q}_3$?

(source: "Introduction to Linear Algebra" by Strang, 5th Ed., 2016)